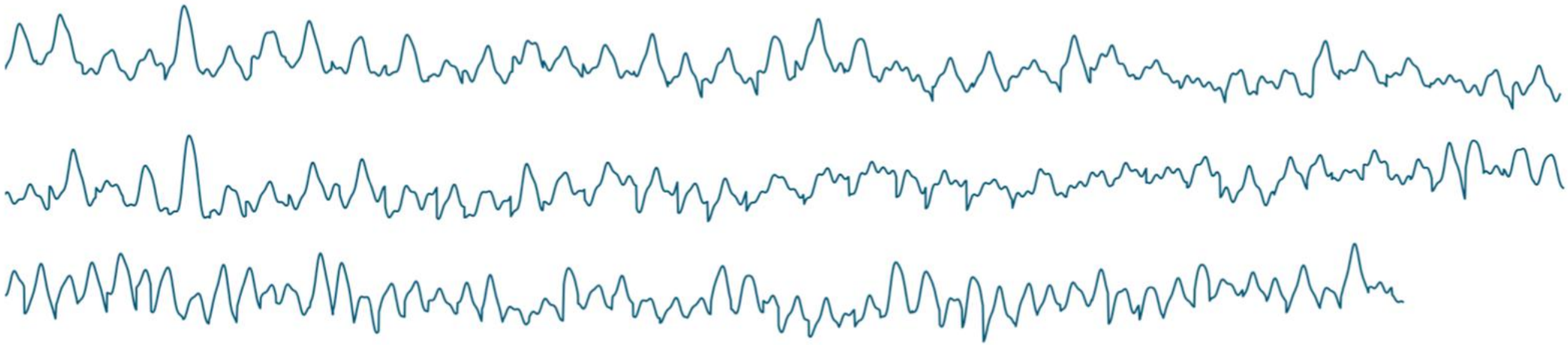
Noéa

Reading water as rhythm, memory and mood

Affective computing for water systems



Hong Kong, 1884–2025: one beat per year. Height is the year's peak daily rainfall; spacing is the time since the previous peak.

MEETING OF WATERS

Data can show that a river is changing. It rarely makes anyone feel it.

DETECT

Stations now measure rainfall, rivers and glaciers at scale.

THE MISSING LAYER

Turning measurement into meaning people can experience.

Noéa works in the middle.

RESPOND

Decisions stall between knowing and acting.

It reads a record the way a cardiologist reads an ECG

01

A long record

Daily rainfall or river discharge, covering decades to centuries.

02

Diagnostics

Peaks and the gaps between them, return periods, flow duration, recession, the seasonal cycle.

03

Two affective values

Valence: pleasant to distressing.

Arousal: calm to activated.

Alongside four physical readouts: size, glow, texture, tempo.

04

XR and sound

Designed to feed immersive and sonic experiences. The output format is still provisional.

Two layers, kept deliberately apart

Science layer

OBJECTIVE

Hong Kong rainfall, 1884–2025: annual peaks, the gaps between them, return periods.

Rhine and alpine rivers, 8 stations: flow duration, recession, the seasonal cycle.

Measures what happened. The measurements are never adjusted for artistic reasons.

Curatorial layer

DESIGNED

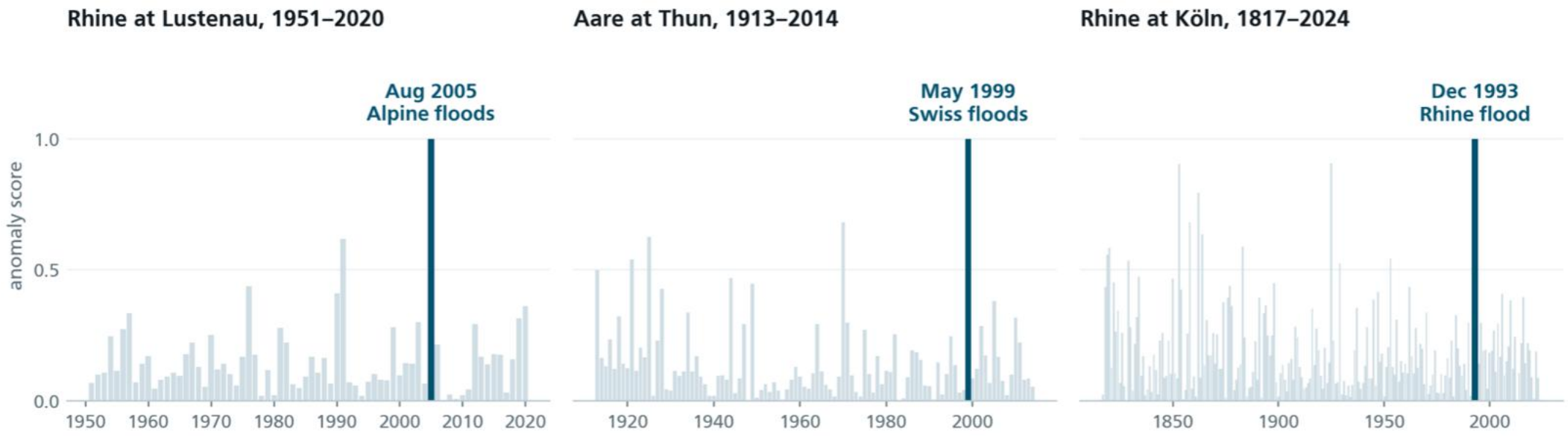
Decides what a drought or a flood should feel like: how alarming, how heavy.

Kept separate from the measurements, so the feeling can be redesigned without changing any of the science.

Its output: valence and arousal.

Machine learning finds unusual years

Its one job so far, unsupervised: learn what a typical year looks like, then score how badly each year fits.



The highlighted bar is each station's highest-scoring year. The model was never told these events existed.

Method: PCA reconstruction error, fit separately per station on its own years. Events: Aug 2005 Alpine floods (22–23 Aug), May 1999 Swiss floods (Thun, 15 May), Christmas 1993 Rhine flood (Köln, 24 Dec).

Built to travel

Each record fits its own model on its own history. Another rainfall or river record can go through the same pipeline.

HONG KONG OBSERVATORY

Hong Kong rainfall

Daily, 1884–2025

One model, fit on 134 years

GRDC, FOEN

Rhine and alpine rivers

Daily, 8 stations

One model per station, 62–208 years

NOT YET RUN

The next record

Another long rainfall or river series.

Tested on two records so far.

The end result is how people respond

The model's output is a score, not a feeling. What a flagged year should feel like comes from people, not from the model.

PERCEPTION

Do people feel what it intends?

The step from anomaly score to emotion is designed by hand. Nobody has tested whether viewers feel it.

UNDERSTANDING AND TRUST

Do they know what a flag means?

A flag is not the whole picture: a drought looks like an ordinary dry year to the model. Do people know what a missing flag does not tell them?

DECISION

Does it help them decide?

Does an XR and sound experience change how decision-makers understand a river, compared with a standard chart?

Next learn the mapping from human responses, and test it with the people who make water decisions.